



Argument Modeling with Aegentic Workflows (An Experimentation)

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INSPIRATION

Ideas and arguments are the primary catalysts for social change and the raw material of human reasoning. A primary example is democracy, which is built on the principle that political power should derive from the consensus - the cumulative reasoning - of the people.

In this exploratory project, we delve into existing methods for modeling reasoning and visualizing arguments derived from conversational data. By leveraging the latest automation technologies, we aim to design constructive workflows for humanities scientists experimenting within artificial ontological environments and diverse narrative schemas.

A deeper understanding of arguments and ideas is the cornerstone of a well-functioning democracy. We could use some of that!

PRINCIPAL CONCEPTS & TOOLS

Phase I: Initial Linguistic Analysis

1. **Distil-Whisper**: Neural transcription of raw audio into high-fidelity text.
2. **SpaCy & ConvoKit**: Syntactic and sociolinguistic feature extraction (e.g., turn-taking, coordination).
3. **System Prompt Library**: Dynamic orchestration of heuristic analysis strategies based on extracted conversation markers.

Phase II: Pattern Recognition

4. **DeepSeek**: Autonomous semantic parsing to identify core entities and relationship triples.
5. **Obsidian**: Ontological synthesis into a linked Markdown ecosystem for dynamic graph visualization.

Phase III: Value Production

6. **Graph Operations**: Algorithmic auditing of the knowledge graph to detect logical inconsistencies.
7. **Testing Universe**: Empirical validation against established conversation datasets for reliability.

WORKFLOWS

Initial Linguistic Analysis

1. **Distil-Whisper**
Neural Speech to text
distillation

2. **Spacy - ConvoKit**
Syntactic Feature Extraction

3. **System Prompt Library**
Heuristic Prompt Orchestration

Pattern Recognition

4. **Deepseek**
Agentic Semantic Parsing

5. **Obsedian**
Dynamic Ontological
Synthesis

Value Production

6. Utilize graph operations to see
inconsistencies

7. **Testing Universe**
Validate workflow on
well studied
conversations

DISCUSSION & CONCLUSIONS

Natural Language Processing is perhaps the most fertile ground for leveraging both the latest automation tools and the vast "deposit" of human knowledge accumulated over the centuries. The potential value of what emerges from the meeting of these two fields is already evident in the sheer scale and promise each holds individually.

If nothing else, this promising direction and the resulting research should be at the forefront of tackling humanity's greatest challenges. If one looks closely, our biggest problems are not technical; we already possess the technology to make significant strides toward solving many of them. Rather, I believe—and this would be a compelling subject for research—that the problems of the modern world are political. While the laws of nature wait patiently for us to discover them, social issues evolve alongside our societies and their history.

REMARKS

We aim for a breadth of potential inputs, though initial testing is anchored in well-studied argumentative paradigms. As this poster contributes to the 'Semantic Annotation for the Ancient World' conference, the Socratic Dialogues remain a non-negotiable cornerstone. Perhaps, even in the era of Artificial Intelligence, we are merely proving **Alfred North Whitehead's** claim that the European philosophical tradition consists of a series of footnotes to Plato. In our case, those footnotes have simply become semantic annotations.

REFERENCES

You can follow the progress of my project and review the academic references in this GitHub repository.

